QoS and Congestion control: explicit, implicit, choking, leaky and token bucket alg., loadshedding

Quality of Service:- In QOS, we try to create an appropriate environment for the traffic.

Flow characteristics:-

- Reliability → Lack of Reliability means Losing or acknowledgment.
- Delay → Source to destination delay.

  S 0 —— 4 D [4]

- Jitter → Variation in Delay.

  P ← 0, 1, 2, 3

  D 4, 5, 6, 7] 4

  4 = 4, 6 = 5, 10 = 8, 15 = 12 → jitter.

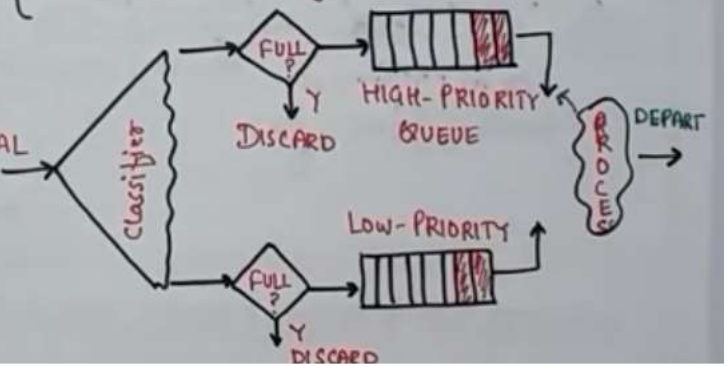
- Bandwidth
  - Different Appl^n Requires diff. Bandwidth.

Techniques to Improve QOS:-

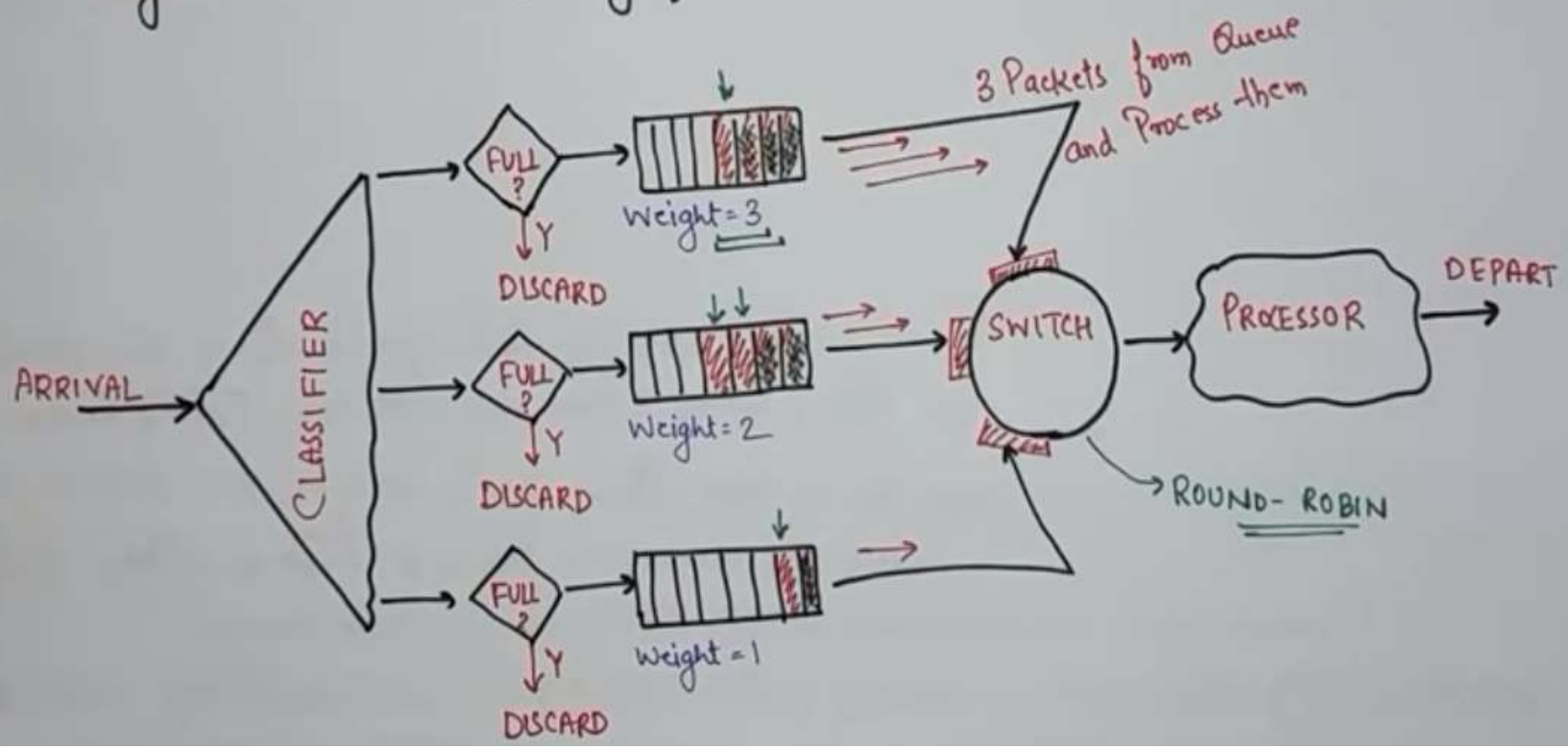
(SCHEDULING)

(i) FIFO Queuing:- FIRST-IN-FIRST-OUT

(ii) Priority Queuing:- STARVATION

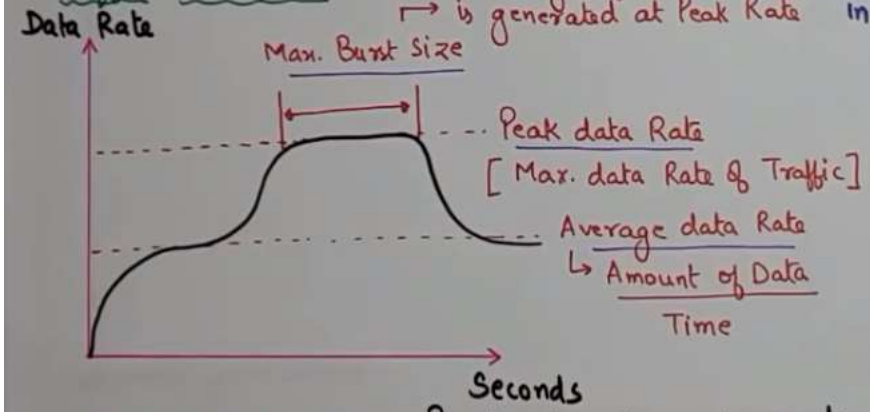


iii) Weighted Fair Queuing:- In this, the Queues are weighted based on the priority of Queue.



Congestion Control:- In this, we try to avoid traffic congestion.

Traffic Descriptors:-



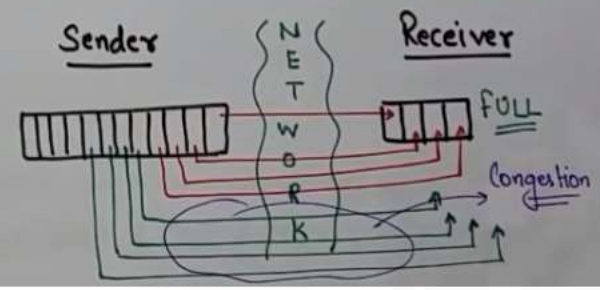
Effective Bandwidth ⇒ Bandwidth that N/W needs to allocate for Traffic Flow.

Congestion:- It may occur if the load on the N/W is greater than the capacity of the N/W. It occurs due to Queues in Routers and switches.

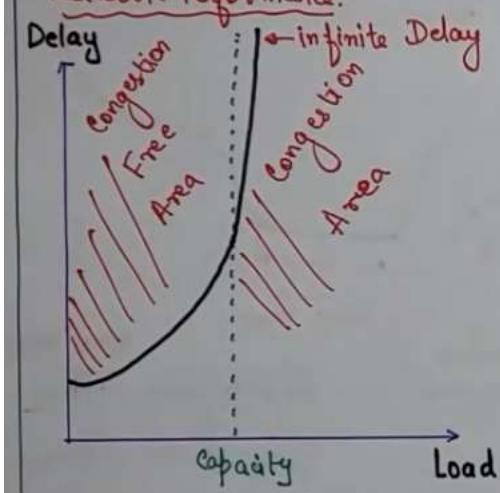
Congestion → LOAD > Capacity

LOAD → no. of Packet sent to N/W

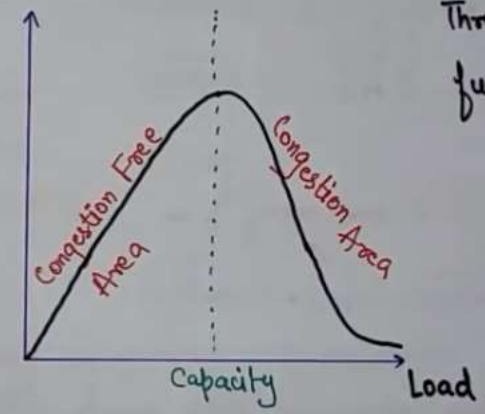
Capacity → No. of Packet a N/W Can Handle.



Network Performance:-

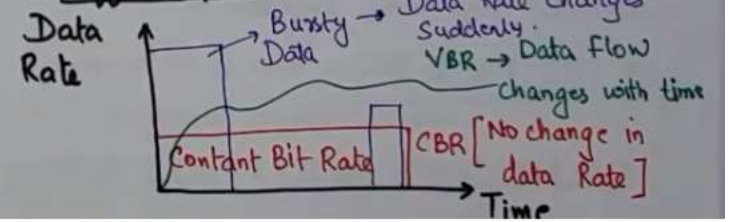


Delay as a func^n of Load



Throughput as a func^n of Load

Traffic Profiles:-

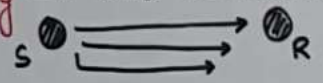


Congestion Control Categories:-

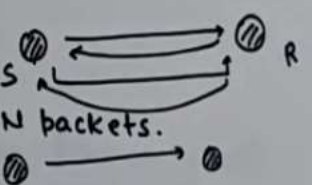
Congestion Control

open-Loop [Prevention]

- Retransmission Policy (TCP) → Good Retransmission.



- Window Policy
  - ↳ Selective Repeat ARQ is better.
- Acknowledgment Policy
  - Receiver will send Ack after a time or N packets.



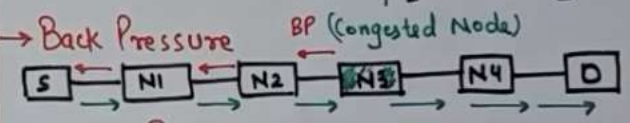
- Discarding Policy
  - ↳ discarding less sensitive packets.

- Admission Policy → Resources are checked first before admitting to the N/W.

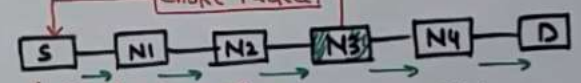
Closed-Loop [Removal]

IMP

- Back Pressure



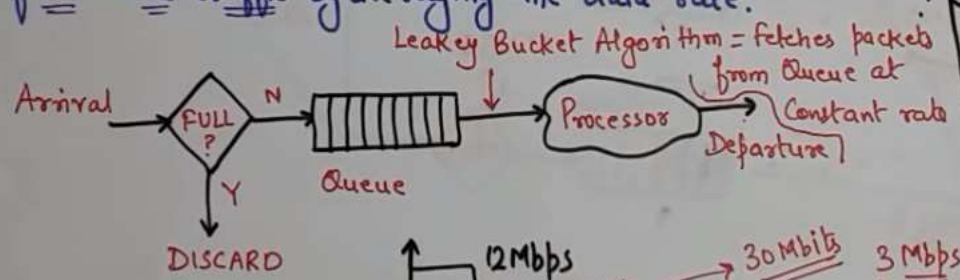
- Choke Packet

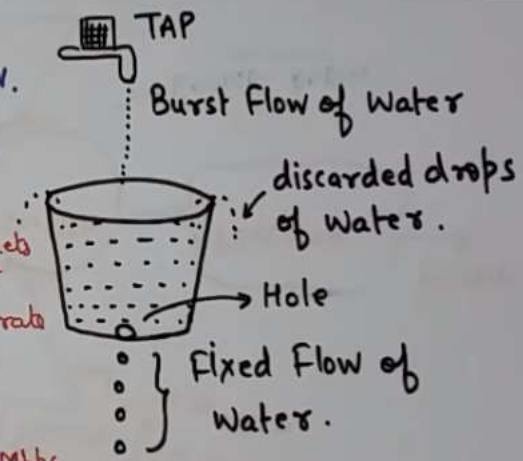


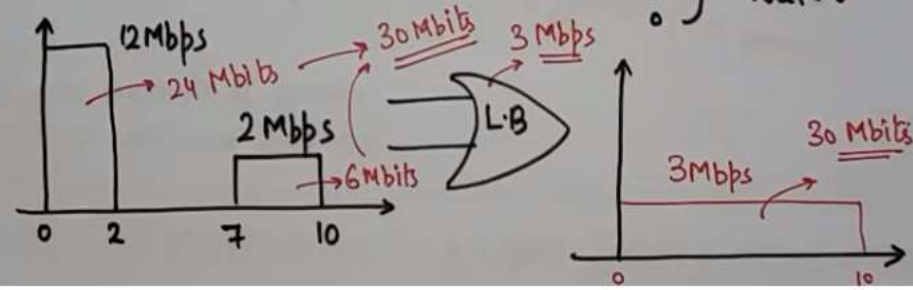
- Implicit Signalling → No comm^n b/w Congested Node with other Nodes/source. Source assumes that Congestion occured.
  - → ACK is Not Rec.
  - → delay in ACK.
- Explicit Signalling
  - ↳ Congested node send a signal to source/destination. Signal is included in the data packet
    - → Backward - Node to source
    - → Forward - Node to Destination

QOS - Traffic Shaping :- It is the mechanism to Control the amount and the rate of the traffic Sent to the N/W.

(i) **Leakey Bucket** :- This algorithm shapes Bursty traffic into fixed-rate traffic by averaging the data rate.

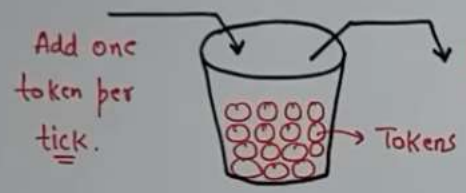


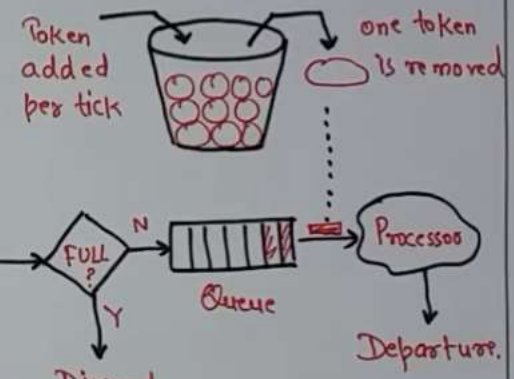




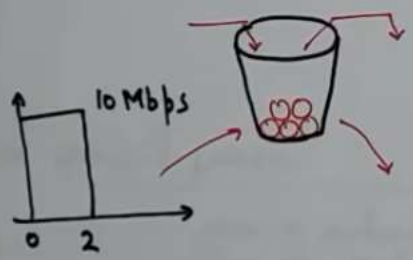
Data Communication and Networking [QOS-4]

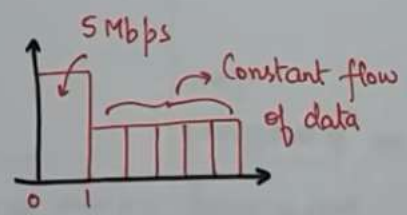
QOS - Token Bucket :- It allows Bursty Traffic at a regulated maximum rate.





IMPLEMENTATION





Q. What is the Difference b/w Leakey Bucket and Token Bucket Algorithm.

TOKEN BUCKET	LEAKEY BUCKET
i) TOKEN Dependent.	i) Token Independent.
ii) If Bucket is full token is discarded but not the packet.	ii) If Bucket is full, then packets are discarded.
iii) Packets Can only transmit when there are enough tokens.	iii) Packets are transmitted Continuously.
iv) Allows Large bursts to be Send at faster rate.	iv) Sends the Packet at a Constant Rate.
v) Saves tokens to Send Large bursts.	v) No Concept of Token.